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# E4 Final Prototype Memo

## Background

Our client works as a Biology Laboratory Technician here at Mudd, where she is responsible for preparing several racks of reagents for use in the freshman Introduction to Biology lab. She typically aliquots these reagents into several 1.7mL microcentrifuge tubes. Before adding reagents to the tubes, our client closes the lids of the tubes and then labels them with stickers to ensure that the correct reagent is added to the correct tube. Since the snap-on lids of the tubes are initially extended at 180 degrees, she closes these tubes 2 at a time by rolling the tubes into an upside-down position before pressing the lids closed against a lab bench. Our client finds that using this process to prepare multiple racks of these tubes is both time-consuming and physically straining on her wrists from the repetitive motion.

## Problem Statement

Through analyzing our client's written statement and in conducting a follow up interview, our team determined the following problem statement:

Our client finds it time consuming and physically straining to manually close the lids of several dozen 1.7 mL microcentrifuge tubes and stick labels to the top before lab each week. She is looking for a solution that reduces strain and the time it takes to prepare the tubes.

The client identified two primary needs: closing tube lids and assisting with sticker application. After ideating multiple design alternatives, our team concluded that these needs must be addressed by developing two separate design prototypes. No single design could effectively perform both functions without compromising performance or functionality, since combining these two needs into one design reduced the sticker application component to a single function for our prototype. Closing the tube lids is what causes the most wrist strain for our client, so we prioritized design developments in this area. The revised problem statement to reflect our decision is as follows:

Our client finds it time consuming and physically straining to manually close the lids of several dozen 1.7 mL microcentrifuge tubes before bio lab each week. She is looking for a solution that reduces strain and the time it takes to prepare the tubes.

# Prototype

The specific question our final prototype aims to answer is: How long does it take to close the lids of eighty 1.7 mL microcentrifuge tubes using our device as compared to our client's extrapolated time?

This question allows us to evaluate how well our prototype fulfills our top ranked objective of minimizing closing timer per tube. This main objective, as determined by our PCC, directly addresses our client's main issue of closing tubes in a timely manner. In conducting our evaluation, we can then identify any time limitations in our mechanism and revise our prototype accordingly. This question can only be answered once we have a high fidelity prototype manufactured with all supporting components, such as extra trays and a tray rack, as a realistic evaluation requires physical testing of our final design. Thus, the success of our prototype, as determined by our question, cannot be answered with background research alone but through physical validation.

# Final Functional Design Justification: Objectives and Constraints

Our objectives listed in order of ranking are as follows:

1. Minimize closing time per tube
2. Maximize number of tubes that can be prepared at once
3. Minimize strain on client's hand
4. Maximize durability of design
5. Minimize cost of each prototype
6. Minimize size of design

| | **Time (seconds)** | **Score** |
|---|---|---|
| **Minimize closing time per tube.** | >3 | 0 |
| | 2.5 - 2.9 | 1 |
| | 2 - 2.4 | 2 |
| | 1.5 - 1.9 | 3 |
| | 1 - 1.4 | 4 |
| | <1 | 5 |

| | **Number of tubes prepared in one round** *One round defined depending on the design alternative* | **Score** |
|---|---|---|
| **Maximize number of tubes that can be prepared at once** | <= 2 tubes | 0 |
| | 3-6 tubes | 1 |
| | 7-10 tubes | 2 |
| | 11-14 tubes | 3 |
| | 15-18 tubes | 4 |
| | > 18 tubes | 5 |

| | **User's perceived comfortability (scale 1-10)** | **Score** |
|---|---|---|
| **Minimize strain on client's hand** | Very uncomfortable ( >= 7) | 0 |
| | Uncomfortable (6-6.9) | 1 |
| | Slightly uncomfortable (5-5.9) | 2 |
| | Slightly more comfortable (4-4.9) | 3 |
| | Comfortable (3-3.9) | 4 |
| | Very comfortable (< 3) | 5 |

| | **Durability** *Qualitative measure determined by client's expertise and research on material used.* | **Score** |
|---|---|---|
| **Maximize durability of device** | Not durable at all | 0 |
| | Somewhat durable | 1 |
| | Durable | 2 |
| | Very durable | 3 |

| | **Cost in dollars** | **Score** |
|---|---|---|
| **Minimize cost of each prototype** | > 100 dollars | 0 |
| | 81-100 dollars | 1 |
| | 61-80 dollars | 2 |
| | 41-60 dollars | 3 |
| | 21-40 dollars | 4 |
| | <= 20 dollars | 5 |

| | **Overall dimensions of design (inches)** *length x width x height* | **Score** |
|---|---|---|
| **Minimize size of design** | > 14 x 10 x 5 (shoebox size) | 0 |
| | 1 inch smaller on one side | 1 |
| | 2 inches smaller on one side | 2 |
| | 3 inches smaller on one side | 3 |
| | 4 inches smaller on one side | 4 |
| | 5 inches smaller on one side | 5 |

Figure 1: Metrics

We evaluated how well our prototype met these objectives as follows:

1: We tested how long it takes to close 80 microfuge tubes with our device, extrapolated our results, and found that it would take 0.658 seconds to close each tube. As a result, our prototype scored a 5 on the metric corresponding to this objective.

2: Since the cap of the microfuge tube extends out 180 degrees, we were limited to closing one row at a time. Although extending the tube tray row would allow for more tubes to be closed at once, we were concerned that it would not allow for the even distribution of force. This is why we chose to make our modified trays to hold 16 tubes. Thus the number of tubes that can be prepared at once is 16 tubes, which scores a 4 on the metric (Figure 1) corresponding to this objective.

3: As an added component to our evaluation of the time objective, we asked each participant who tested our device to rate the exertion they experienced on a scale of 1-10. The average exertion rating was found to be 1.8. As a result, our prototype scored a 5 on the metric corresponding to this objective.

4: We built the base and supports of our final prototype out of aluminum. Based on our research we found that aluminum is a durable metal that can withstand the chemicals that our prototype will encounter in the biology lab. The lego base was printed out of carbon fiber, which is a durable material with a high strength-to-weight ratio. We printed 1 tray out of carbon fiber and 4 other trays plus the convex board out of thin layered PLA and higher infill density, since the thinner layers and higher infill give the prints more structural integrity. PLA is less durable than carbon fiber, however, we wanted to give our client different price options to see if it is truly necessary to print everything out of carbon fiber. The carbon fiber print costs $0.21 per gram, while the PLA print is free.

5: The cost to produce the final prototype was $61.40. This would score a 2 on the metric corresponding to this objective, which is not ideal. However, our team made design choices that prioritized durability over price. For example, we hope that the durability of the carbon fiber will allow our parts to last longer, reducing our cost in the long term.

6: When measured with a ruler the dimensions of our device in inches were 10in x 4.1in x 3.5in. This would score a 5 on the metric corresponding to this objective, and make the prototype easily transportable by our client.

Our constraints are as follows:

1. Must not break tubes
2. Must close 68 tubes in less than 3.5 minutes
3. Must cost under $30
4. Must not damage user's gloves
5. Must withstand sterilization with ethanol

Verifying that our prototype met these constraints:

1: None of the tubes broke during testing, which involved repeated closing and opening of tubes. As a result, our prototype satisfies this constraint.

2: We evaluated how long it takes to close 80 microfuge tubes with our device, and then used that time value to mathematically extrapolate how long it would take to close 68 tubes. Based on this method, we found that it would take 44.8 seconds to close 68 tubes, which is significantly less than the 210 seconds it took our client to close 68 tubes. As a result, our prototype satisfies this constraint.

3: The cost to produce the final prototype was $61.40 which is double what our original constraint was. We are aware that we made some design choices that prioritized durability over price, since we wanted to ensure that our prototype did not need to be replaced or repaired in the near future. A large part of our decision to do this was to give our client to test out different materials over a long period of time, which we were not able to do ourselves due to our time constraints on this project. If we were to go with our original design plan, we estimate the cost of our prototype to be around 28 dollars, which would have satisfied our constraint.

4: We rounded off the corners of our aluminum pieces to prevent loose materials such as latex gloves or lab coat sleeves from getting caught and damaged in our prototype. As a result, our prototype satisfies this constraint.

5: Upon research, we found that aluminum, carbon fiber, and PLA can withstand sterilization with ethanol. As a result, our prototype satisfies this constraint. While we did have concerns about how PLA would react with ethanol over a long period of time, our decision to provide our client with multiple different parts made of different materials allows us to follow up with our client after a period of time to see how well our materials hold up over time. However, based on the testing we were able to perform within our time constraint, we found no reason to be concerned about any of our PLA parts.

## Final Functional Design Justification: Material Choices

The materials we selected for our final prototype are as follows:

1. Aluminum
2. Stainless Steel
3. Carbon fiber nylon filament
4. PLA filament

Justification for selection of material:

1: We selected aluminum for the base and supports to maximize the durability of our design. Not only does aluminum have a high strength-to-weight ratio, but it can withstand ethanol use, ensuring that it maintains its structural integrity after repeated use. We also anticipate that the base will come into contact with a variety of chemicals as it is placed on different lab benches, and based on our research of the common chemicals found in the biology lab, we believe that our aluminum base will be able to withstand all the conditions it is placed in.

2: We selected stainless steel for the dowel pin, bearings, bolts, nuts, and washers to maximize the durability of our design. While stainless steel is a little more expensive, we wanted to ensure that the dynamic parts of our prototype are made of a high quality material that can withstand ethanol sterilization and the wear and tear of repeated motion. In addition, we chose stainless steel because of its resistance to rusting, which is really important for our bearings and dowel pins since we do not want them to inhibit the rotation of our device.

3: We selected carbon fiber nylon filament for the LEGO board and 1 tray to maximize the durability of our design. Carbon fiber has a high strength-to-weight ratio, making it a more durable material. We also found through testing that carbon fiber can withstand ethanol sterilization and does not expand or swell when exposed to high amounts of liquid, which is important for our LEGO board since it needs to maintain its structure to fit into other parts.

4: We printed 4 additional trays and our press board out of PLA. Thinner print layers and higher infill density give the prints increased structural integrity. While we are aware that PLA is less durable than carbon fiber, we wanted to give our client different price options to see if it is truly necessary to print everything out of carbon fiber. Since we found no immediate concerns about PLA holding up in a biology laboratory setting, we hope that our client is able to test these parts over a longer period of time and help us with our evaluation about the durability of PLA when exposed to ethanol over time.

## Prototype Functionality

Our primary functions are as follows:

1. Distributes uniform pressure on tube caps
2. Alleviates wrist strain during lid closure
3. Maintains structural integrity after repeated use
4. Maintains tube orientation after inserted

Verifying that our prototype met these functions:

1: We know that uniform pressure is distributed on all the tubes, since all the lids consistently close upon applying force with 2 hands onto the board.

2: When performing our final prototype evaluation, we asked each participant who tested our device to rate the exertion they experienced on a scale of 1-10. The average exertion rating was found to be 1.8, which scores a 5 on the metric corresponding to minimizing strain. Using these results, we hypothesize that over a long period of time, our client will feel less wrist strain using our device than what they currently feel.

3: We built our final prototype out of materials that we researched to be both chemically and mechanically durable. In addition, after repeated testing using our device, we did not see any component break or deform. We are aware that structural integrity is best evaluated after long term repeated use, and will check in with our client after one semester of use to see how well our design held up after repeated use.

4: The device requires the tubes to be in the right orientation to effectively close the lids. Since our device is able to consistently close all of the tubes, we are confident that our tray is holding our tubes in the correct orientation during the closing process.

At our last client meeting, we did not have our final prototype fully assembled, so we were advised by Prof Trank to show our client all the components of our final prototype and explain how each part corresponded to an aspect of our fully functional medium resolution prototype. She approved of all the components and expressed excitement at the prospect of receiving the final prototype in the near future. She also commented on how she liked the professional and neat aesthetic of our device.

# Evaluation Plan

Evaluation Question: How long does it take to close the lids of 80 1.7 mL microcentrifuge tubes with our device compared to our client's current method?

Evaluation Sub-question: Does using the prototype reduce hand strain for the client compared with manual closing (closing two at once just using one's own hands), after closing the 80 tubes?

Initial Hypothesis: When tubes are placed in the correct orientation (see Figure 2), a single user will be able to close all five racks (80 tubes total) in approximately 1 minute using the prototype.

Experimental Variables: The independent variable is the time it takes the participants to close 80 tubes using the prototype. Each person will repeat this process for 3 trials. The dependent variables are the total time required to close all the tubes in the five racks and the perceived hand strain of the process. The control variable is that each participant will be given the exact same setup and will close exactly 80 tubes.

Experimental Data Collection: We recorded how long it took each participant to close the lids of 80 tubes over the course of three trials, calculating each participant's average. The standard deviation of each participant's results were also calculated in order to evaluate whether there was a learning curve associated with the task as the participant gained familiarity with the prototype, potentially skewing the data. We further recorded qualitative data about the hand strain the participant experienced while performing the test, in which each participant was asked to quantify their wrist strain on a scale of 1 to 10 after all 3 trials. All collected data can be viewed in Table 1 below.

Evaluation Plan:

1. Prepare set up in accordance with Figure 2.
2. Tell participants to place a rack onto the prototype, close all the lids with a press board, and repeat until all the tube lids on each rack are closed and placed in the tray holder as shown on the left in Figure 2.
3. Start an iphone timer when the first rack is placed into the prototype, and end the time when the final rack with (with all the tubes closed) is placed into the tray holder.
4. Note the time in Table 1.
5. Repeat steps 1-4 for two more trials for the same participant.
6. Calculate the average and standard deviation for each participant across the 3 trials.
7. Ask the participant to rate their exertion (strain) from 1-10 and note the value in Table 1.
8. Repeat steps 1-6 for a total of 6 participants.
9. Calculate the average of the averages of the 6 testers and rated exertion.

Evaluation success: Our quantitative success was defined as obtaining an overall average closing time of under a minute across all test participants. Our qualitative success was defined as obtaining an overall average of 3 for perceived hand strain across all test participants.

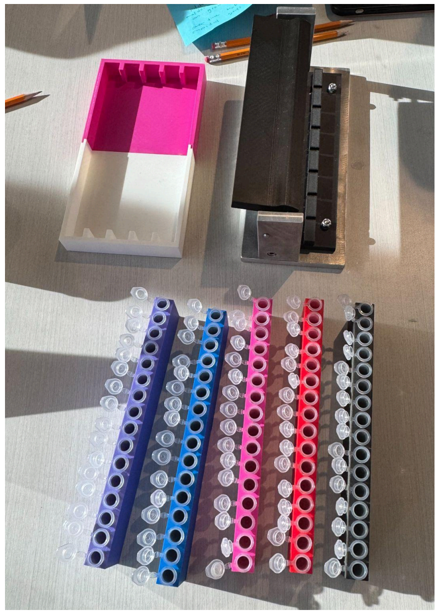

Figure 2: Setup for Evaluation

## Evaluation Results

| | Amy | Heidi | Charlotte | Helen | Nicole | Ellie | |
|---|---|---|---|---|---|---|---|
| Participant | 1 | 2 | 3 | 4 | 5 | 6 | |
| Trial 1 (s) | 68.46 | 88.86 | 44.02 | 62.56 | 61.96 | 46.9 | |
| Trial 2 (s) | 65.24 | 53.99 | 42.37 | 45.85 | 47.04 | 36.44 | |
| Trial 3 (s) | 65.08 | 46.54 | 47.34 | 45.44 | 47.27 | 32.35 | Average for 80 Tubes (s) |
| Average | 66.26 | 63.13 | 44.577 | 51.283 | 52.09 | 38.563 | 52.65 |
| SDev | 1.907 | 22.592 | 2.531 | 9.768 | 8.548 | 7.503 | Average Rated Exertion |
| Rated Exertion (1-10) | 4 | 2 | 1 | 1 | 2 | 1 | 1.83 |

Table 1: Final prototype evaluation data.

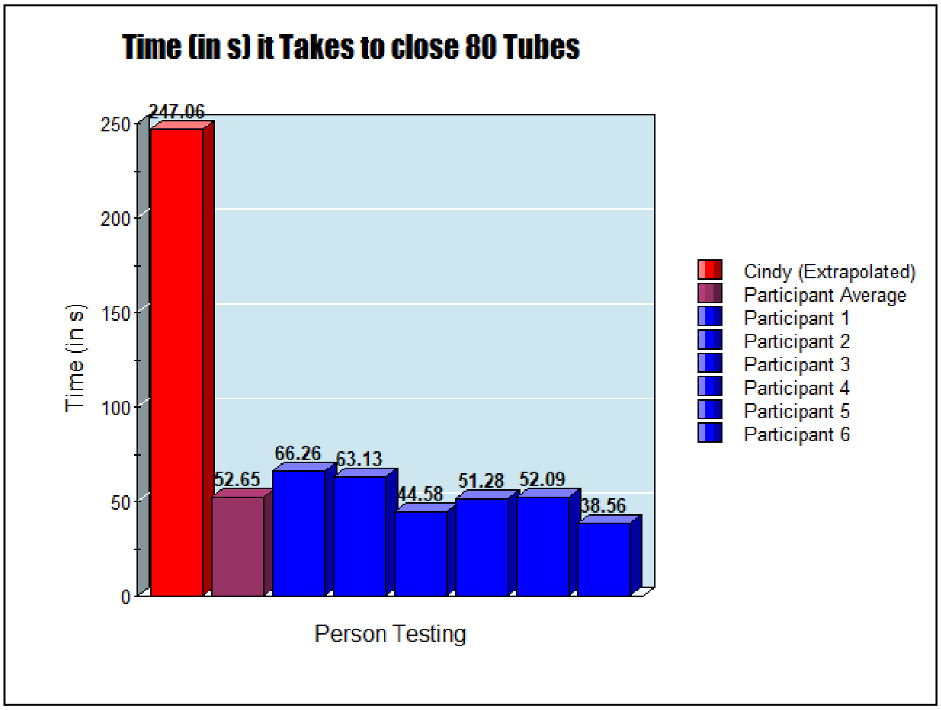


Figure 3: Graphic summary of evaluation results.

Table 1 presents the data collected from the evaluation plan. The average time required to close 80 tubes across all six testers was approximately 52.65 seconds, while the average rated exertion was 1.83. Based on these results, our prototype met the evaluation criteria for a successful design. Figure 3 summarizes our findings and includes our client's original tube-closing time as a reference. From this comparison, we observe that the prototype enabled tube closure at a rate approximately 4.7 times faster than the client's original method.

We predict that one source of uncertainty in our data comes from a learning curve that may have occurred after participants completed multiple trials of the evaluation test. As shown in Table 1, the standard deviation is noticeably high for Heidi and moderately high for Helen, Nicole, and Ellie. This may have been due to participants learning through their own trials and as a result becoming more skilled at placing the racks and closing the tubes over time. For future tests, we could minimize this uncertainty by allowing participants to practice with the prototype before performing recorded trials or by taking more trials to get a more accurate performance average.

Another possible source of uncertainty was that some of the participants had trouble placing and removing the tray (specifically the purple tray). They noted that this was due to a few structure abnormalities in the tray that made it difficult to place the tray on and remove the tray from the lego base. This systematic error may have increased the times for all participants. In future tests, we plan on either replacing that tray all together or filing it down so that it fits better into the Lego base.

Lastly, there was a source or random error across all trials that came from manually starting and stopping the timer for each trial. Since the timer is started and stopped by human reaction time, it is impossible to be perfectly precise in timing each participant. However, since this is a source of random error, we do not believe it significantly impacted our results.

## Client Impact

While our client was not able to evaluate the final prototype at our last client meeting, we were able to show her all the components of the prototype. She had no concerns about our material or design choices, and was satisfied with the projected finished final prototype. During this client meeting, she also mentioned a few additional design requests, including a device to open all the tubes while they are in the tray and a way of organizing multiple trays. We used this feedback to add a few extra components to our design, including a separate tube opening mechanism and a tray holder for organization. Although these secondary designs were not part of our original problem statement, we hope that our solutions are able to satisfy the client's additional needs to create a better tube closing experience.

Our design keeps the microcentrifuge tubes upright during the closing and opening process, which additionally allows the client to use the device when tubes have reagents inside. As a whole, our final prototype meets the client's top-priority needs by making the tube-closing process faster and reducing wrist strain.